
Answer Keys – All Worksheets

Note: This section contains answers for all 10 worksheets. Teachers may detach or print this section separately.

Worksheet 1: Chapter 1: Patterns in Mathematics**Answer Key****Answer Key – Worksheet 1: Chapter 1: Patterns in Mathematics**

1. 12, 14 2. 21, 28 3. Add 5 each time (multiples of 5) 4. 16 5. 1, 4, 9, 16, 25
6. 3, 10, 17, 24, 31, 38 7. 21 8. 13, 21, 34
9. $6^2 = 36$ 10. 125
11. 1234321 12. $T_{10} = 55$; $T_{12} = 78$ 13. 100, 89, 78, 67, 56, 45
14. Multiply by 2; 8th term $= 3 \times 2^7 = 384$ 15. $T_3 + T_4 = 6 + 10 = 16 = 4^2$ ✓
16. $16 + 9 + 4 + 1 = 30$ squares 17. $\frac{10 \times 9}{2} = 45$ handshakes
18. 2, 5, 7, 12, 19, 31, 50, 81
19. Row 8: $10 + 7 \times 3 = 31$ seats; Total $= 8 \times 10 + 3(0 + 1 + 2 + \dots + 7) = 80 + 84 = 164$ seats
20. Week 8: $50 \times 2^7 = 6400$; Total $= 50 \times (2^8 - 1) = 12750$

Worksheet 2: Chapter 2: Lines and Angles**Answer Key****Answer Key – Worksheet 2: Chapter 2: Lines and Angles**

1. Lines that never meet; railway tracks / ruled notebook lines
2. 90 3. Obtuse 4. 110 5. 48
6. 55 7. 120, 60, 120, 60 8. 90
9. Right angle 10. 75, 105, 75, 105
11. $x + 2x + 3x = 180$; $x = 30$
12. 30, 60, 90; Right-angled triangle
13. 75 14. 180; other angle = 115
15. 120; Obtuse
16. $3x + 10 = 5x - 30$; $x = 20$; angle = 70
17. $2k + 3k + 4k = 180$; $k = 20$; angles: 40, 60, 80; No angle > 90 (all acute)
18. Third angle $= 180 - 50 - 70 = 60$; Acute triangle
19. Angle with wall = 20; 70 = acute, 20 = acute
20. Third angle $= 180 - 55 - 90 = 35$

Worksheet 3: Chapter 3: Number Play

Answer Key**Answer Key – Worksheet 3: Chapter 3: Number Play**

1. Last digit is 0 or 5 2. Yes; last digit is 0 3. $5 + 3 + 1 = 9$; divisible by 3 ✓
4. 11 5. 1, 2, 3, 4, 6, 8, 12, 16, 24, 48
6. $3 + 6 + 7 + 2 = 18$; By 2: Yes; By 3: Yes; By 4: $72 \div 4 = 18$ Yes; By 6: Yes; By 9: Yes
7. $36 = 2^2 \times 3^2$; $84 = 2^2 \times 3 \times 7$; HCF = $2^2 \times 3 = 12$
8. LCM(12, 15, 20) = 60 9. No; 12 is divisible by 4 and 6 but not 24
10. 53, 59, 61, 67, 71, 73, 79
11. $56 = 2^3 \times 7$; $98 = 2 \times 7^2$; HCF = 14; LCM = 392; $14 \times 392 = 5488 = 56 \times 98$ ✓
12. $(5 + 9 + 8) - (4 + 7) = 22 - 11 = 11$; divisible by 11 ✓
13. LCM(2, 3, 4, 5, 6) = 60 14. Other = $\frac{12 \times 360}{60} = 72$
15. $120 = 2^3 \times 3 \times 5$
16. $99999 \div 36 = 2777.75$; $2777 \times 36 = 99972$
17. LCM(12, 18, 24) = 72 min; Next ring at 9:12 AM; Between 8–12 (240 min): $240 \div 72 = 3$ remainder 24; they ring together 4 times (at 0, 72, 144, 216 min)
18. LCM(5, 7, 9) = 315; $N = 315 + 3 = 318$
19. HCF(252, 336) = 84; Row size = 84; Boys: 3 rows, Girls: 4 rows
20. LCM(6, 8, 12) = 24 m

Worksheet 4: Chapter 4: Data Handling and Presentation**Answer Key****Answer Key – Worksheet 4: Chapter 4: Data Handling and Presentation**

1. Mean = $30 \div 5 = 6$ 2. Mode = 3 3. Sorted: 3, 5, 7, 9, 12; Median = 7
4. Three groups of 4 + two singles =  5. Pictograph
6. Mean = $(45 + 62 + 58 + 71 + 45 + 89 + 62 + 45) / 8 = 477 / 8 = 59.6$; Mode = 45; Sorted: 45, 45, 45, 58, 62, 62, 71, 89; Median = $(58 + 62) / 2 = 60$
7. Red: 4, Blue: 3, Green: 2, Yellow: 1
8. Average = $(45 + 60 + 30 + 75 + 50) / 5 = 52$ runs
9. Maths most popular 10. Sum = $5 \times 20 = 100$
11. Sum = 205; Mean = 20.5; Mode = 20; Sorted: 18, 18, 19, 20, 20, 20, 21, 22, 23, 24; Median = 20
12. Mean = $120 / 6 = 20$; Modes = 18 and 22 (bimodal)
13. Sum of all 6 = $15 \times 6 = 90$; Sixth = $90 - (12 + 18 + 14 + 16 + 10) = 90 - 70 = 20$
14. Sorted: 19, 19, 23, 28, 28, 28, 35, 35, 41; Mean = $256 / 9 \approx 28.4$; Mode = 28; Median = 28; Range = 22
15. Teacher Verification Required
16. New sum = $20 \times 72 + (84 - 48) = 1440 + 36 = 1476$; New mean = $1476 / 20 = 73.8$
17. Median is 4th value; x can be any value between 21 and 25 (e.g., 23, 24, 25)
18. Priya is correct; e.g., for skewed salary data, median is more representative than mean
19. Mean = $287 / 6 \approx 47.8$ mm; Months above avg: Apr, May, Jun
20. Teacher Verification Required

Worksheet 5: Chapter 5: Prime Time**Answer Key****Answer Key – Worksheet 5: Chapter 5: Prime Time**

1. 2,3,5,7,11,13,17,19,23,29 2. No; 1 has only one factor
3. 2 4. $\text{HCF}(8,15)=1$; Yes, coprime 5. Four (11,13,17,19)
6. 2,3,5,7,11,13,17,19,23,29,31,37,41,43,47 (Teacher Verification Required for sieve)
7. $36 = 5 + 31 = 7 + 29 = 13 + 23$ (any two valid) 8. $\text{HCF}(25,36)=1$; Yes coprime
9. (3,5), (5,7), (11,13) (accept any three) 10. $28=11+17$; $36=7+29$; $50=3+47$
11. Exactly 2 (1 and itself) 12. $4(=2^2)$, $9(=3^2)$, $25(=5^2)$; squares of primes
13. Yes (all primes > 5 are odd); yes (must end in 1,3,7, or 9)
14. $120 = 2^3 \times 3 \times 5$ 15. 96: factors 1,2,3,4,6,8,12,16,24,32,48,96 (12 factors)
16. Proper factors of 28: 1,2,4,7,14; sum=28 ✓; 28 is perfect
17. 3,13,23,43,53,73,83 — **7 primes**
18. pq : 4 factors; p^2q : 6 factors; In general $p^a \times q^b$ has $(a+1)(b+1)$ factors
19. Factor 77: try $7 \times 11=77$; For 143: $11 \times 13=143$
20. $120 = 2^3 \times 3 \times 5$; prime divisors: 2, 3, 5

Worksheet 6: Chapter 6: Perimeter and Area**Answer Key****Answer Key – Worksheet 6: Chapter 6: Perimeter and Area**

1. 36 cm 2. 46 cm 3. 30 cm 4. 44 cm 5. 30 cm
6. 270 m 7. $r = \frac{88 \times 7}{2 \times 22} = 14$ cm 8. 30 cm 9. 17 cm
10. $\pi r + 2r = \frac{22}{7} \times 7 + 14 = 22 + 14 = 36$ cm
11. $2(10 + 8) = 36$ m (outer); notch doesn't change perimeter: $P = 36$ m
12. $2(l + b) = 240$; $l = 2b$; $b = 40$ m, $l = 80$ m
13. $4 \times 2 \times \frac{22}{7} \times 35 = 880$ m
14. Same perimeter; square side= 22 cm, Area= 484 cm^2 ; circle radius= 14 cm, Area= 616 cm^2 ; Circle has larger area
15. Side = $72 \div 6 = 12$ cm; circumscribed radius = 12 cm
16. Hypotenuse= $\sqrt{24^2 + 10^2} = 26$ m; Perimeter= 60 m
17. $2(l + b) = 60$; $l - b = 6$; $l = 18$, $b = 12$ m; Area= 216 m^2
18. Triangle has side = 8 cm (between centres); perimeter of curved region: $3 \times (\text{arc}) = 3 \times \frac{60}{360} \times 2\pi \times 4 = 3 \times \frac{4\pi}{3} = 4\pi \approx 12.6$ cm
19. $2 \times 100 + 2 \times \pi \times 35 = 200 + 220 = 420$ m
20. Pool perimeter= 150 m; Outer: $(50 + 4) \times 2 + (25 + 4) \times 2 = 108 + 58 = 166$ m; Extra= 16 m

Worksheet 7: Chapter 7: Fractions**Answer Key**

Answer Key – Worksheet 7: Chapter 7: Fractions

1. Proper; Improper; Mixed; Improper (equal to 1)
2. $\frac{14}{3}$ 3. $3\frac{7}{8}$ 4. $\frac{6}{10}, \frac{9}{15}$ (any valid pair)
5. HCF(36,48)=12; $\frac{3}{4}$
6. LCM(8,12)=24; $\frac{15}{24}$ vs $\frac{14}{24}$; $\frac{5}{8} > \frac{7}{12}$
7. LCD=12; $\frac{10}{12}, \frac{10}{12}, \frac{9}{12}, \frac{7}{12}$; Descending: $\frac{5}{6} = \frac{3}{2} > \frac{2}{3} > \frac{7}{12}$
8. Girls=15; Boys=25 9. HCF(72,108)=36; $\frac{2}{3}$
10. $\frac{3}{8}$ (any fraction between $\frac{1}{4}$ and $\frac{1}{2}$)
11. $\frac{20+21}{24} = \frac{41}{24} = 1\frac{17}{24}$
12. $3\frac{3}{12} - 1\frac{10}{12} = 1\frac{5}{12}$
13. $\frac{4 \times 15}{5 \times 16} = \frac{60}{80} = \frac{3}{4}$
14. $\frac{7}{8} \times \frac{32}{21} = \frac{224}{168} = \frac{4}{3} = 1\frac{1}{3}$
15. Other = $\frac{5}{12} \div \frac{5}{8} = \frac{5}{12} \times \frac{8}{5} = \frac{2}{3}$
16. Fraction spent = $\frac{1}{4} + \frac{1}{3} + \frac{1}{6} = \frac{3+4+2}{12} = \frac{9}{12} = \frac{3}{4}$; Saved = $\frac{1}{4}$; Income = 6000
17. $\frac{4}{5} - \frac{3}{5} = \frac{1}{5}$ of tank = 45 L; Capacity = 225 L
18. Per person: $\frac{9/4}{6} = \frac{3}{8}$ cup; For 10: $\frac{15}{4} = 3\frac{3}{4}$ cups; For 4: $\frac{3}{2} = 1\frac{1}{2}$ cups
19. Cricket: 500; Football: 300; Badminton: 200; None: $1200 - 1000 = 200$
20. Sold Mon: 36 kg; Remainder: 24 kg; Sold Tue: 6 kg; Left: 18 kg

Worksheet 8: Chapter 8: Playing with Constructions**Answer Key****Answer Key – Worksheet 8: Chapter 8: Playing with Constructions**

1. Compass and ruler (straight edge)
2. It cuts the segment into two equal halves at 90
3. Two equal halves
4. Teacher Verification Required
5. 60; Construct by drawing a circle, then marking off a chord equal to the radius
- 6–15. Teacher Verification Required for all constructions
16. Yes: bisect 90 to get 45; Teacher Verification Required
17. Circumcentre; it is equidistant from all three vertices and is the centre of the circumscribed circle
18. Teacher Verification Required; Each side of a regular hexagon equals the radius because the hexagon is made of 6 equilateral triangles

19. Draw both diagonals of the rectangle; their intersection is the centre
20. Mark two points for one wall, find the perpendicular bisector to get the right-angle direction, measure equal lengths for the adjacent wall

Worksheet 9: Chapter 9: Symmetry

Answer Key

Answer Key – Worksheet 9: Chapter 9: Symmetry

1. 3 2. 4 3. Yes; 2 lines (horizontal and vertical)
4. 120 5. All three (S, N, Z have order-2 rotational symmetry and no line symmetry)
6. Circle: infinite; Rectangle: 2; Regular pentagon: 5; Regular hexagon: 6
7. 8 sides (regular octagon); order 8 8. Yes; 2 lines of symmetry (vertical and horizontal)
9. Yes; order of rotational symmetry = 3 10. Taj Mahal, Red Fort (accept any valid answer)
11. n lines of symmetry; order n
12. A, T, U, V, W, Y (any 4 correct)
13. No line of symmetry; Yes, rotational symmetry of order 2 (180°)
14. 60, 120, 180, 240, 300, 360
15. Teacher Verification Required
16. Rotational: 90, 180, 270, 360; Lines: 4 (2 through midpoints of opposite sides + 2 diagonals)
17. Yes; e.g., letter S or Z; rotates to look same at 180° but has no mirror symmetry
18. Teacher Verification Required (rotate and check)
19. Teacher Verification Required
20. 3-blade fan: order 3; angles 120, 240, 360; 4-blade: order 4; angles 90, 180, 270, 360

Worksheet 10: Chapter 10: The Other Side of Zero (Integers)

Answer Key

Answer Key – Worksheet 10: Chapter 10: The Other Side of Zero (Integers)

1. $-7, +12, 0, +25$ 2. $-3 > -8$ 3. $-9, -5, -1, 0, 3, 7$
4. $|-15| = 15$; $|+8| = 8$ 5. -15 ; $+200$; -200
6. -8 7. -4 8. -3 9. -30 10. $+6$
11. $+1$ 12. $+56$ 13. -4
14. $a + b = 2$; $a - b = -8$; $a \times b = -15$; $a \div b = -\frac{3}{5}$
15. $+6 \times -4 = -24$; $-24 \div -4 = +6$; $-9 \times -3 = +27$; $+27 \div -3 = -9$; $-5 \times +7 = -35$; $-35 \div +7 = -5$
16. Noon: -2 ; Evening: $-2 - 8 = -10$; Midnight: $-10 + 5 = -5^\circ\text{C}$
17. Start: -200 ; Rise 75: -125 ; Dive 120: -245 m

18. Sum = 0 (each negative cancels with its positive; pairs $(-10+10), (-9+9) \dots = 0$)
19. Net = $350 - 120 - 80 + 200 = 350$ profit
20. Sum = $-3 - 1 + 2 + 5 - 2 + 4 + 1 = 6$; Mean = $6 \div 7 \approx 0.86^\circ\text{C}$; Below average:
Mon(-3), Tue(-1), Fri(-2) — 3 days

Teacher Notes

Chapter 1: Patterns in Mathematics

Learning Outcomes: Students identify number sequences (counting, square, triangular, Fibonacci, cube); describe rules; extend patterns; and relate shape patterns to number sequences.

Common Misconceptions: Students often guess next terms without identifying a rule. They may confuse adding a constant with multiplying. Emphasise writing the rule first.

Activities: Pascal’s triangle; dot patterns; staircase patterns with blocks; Fibonacci in sunflower seeds.

Remediation: Begin with visual patterns (shapes, tiles) before moving to numeric. Always ask: “*What is the rule?*” before finding the next term.

Chapter 2: Lines and Angles

Learning Outcomes: Students classify angles; apply properties of parallel lines (alternate, corresponding, co-interior); find missing angles in triangles and quadrilaterals.

Common Misconceptions: Students confuse complementary (90°) and supplementary (180°). They may misidentify alternate vs corresponding angles. Use colour-coding on diagrams.

Activities: Angle scavenger hunt in the classroom; clock angle activities; straw-and-pin models of parallel lines and transversals.

Remediation: Always draw and label diagrams. Create an “angle properties chart” for student reference.

Chapter 3: Number Play

Learning Outcomes: Students apply divisibility rules; find HCF and LCM using prime factorisation; identify primes; solve real-life sharing/scheduling problems.

Common Misconceptions: Students confuse HCF (for dividing/sharing) with LCM (for scheduling/combining). They may think 1 is prime.

Activities: Sieve of Eratosthenes on a 100-chart; factor tree art; bus-timetable LCM puzzles.

Remediation: Use mnemonics: H = Highest $\rightarrow$ smallest; L = Lowest $\rightarrow$ largest (in terms of the answer compared to the inputs).

Chapter 4: Data Handling and Presentation

Learning Outcomes: Students organise data using tally tables; calculate mean, mode, and median; draw and interpret bar graphs and pictographs.

Common Misconceptions: Students add data values and forget to divide for mean. They may pick any central value as median without sorting.

Activities: Classroom surveys; weather data tracking; cricket scorecard analysis.

Remediation: Always sort data before finding median. Use physical objects (counters) to demonstrate finding the middle value.

Chapter 5: Prime Time

Learning Outcomes: Students identify primes; understand coprimality; apply the Fundamental Theorem of Arithmetic; connect primes to cryptography and real life.

Common Misconceptions: Students may test only small primes when checking primality. Teach: only test primes up to $\sqrt{N}$.

Activities: Sieve of Eratosthenes; prime number art; twin prime investigation.

Remediation: Create a prime number reference card (1–100). Play “Factor It!” card games.

Chapter 6: Perimeter and Area

Learning Outcomes: Students compute perimeters and areas of squares, rectangles, triangles, parallelograms, and circles; solve composite figure problems.

Common Misconceptions: Students confuse perimeter (boundary, linear units) with area (surface, square units). They may use wrong formula for circle (using diameter instead of radius).

Activities: Measure and tile the classroom floor; fencing problems on the school ground; circumference with thread and cylinder.

Remediation: Always write units: cm vs cm^2 . Draw and shade the region for area; trace the boundary for perimeter.

Chapter 7: Fractions

Learning Outcomes: Students classify fractions; find equivalent fractions; compare, add, subtract, multiply, and divide fractions; solve word problems.

Common Misconceptions: Students add numerators and denominators separately. They may invert the wrong fraction when dividing.

Activities: Fraction strips; pizza/chapati cutting models; equivalent fraction matching games.

Remediation: For addition, always find LCM of denominators first. For division, always use “keep-change-flip.”

Chapter 8: Playing with Constructions

Learning Outcomes: Students construct perpendicular bisectors, angle bisectors, specific angles, and triangles using compass and ruler.

Common Misconceptions: Students press too hard with the compass, slipping and making inaccurate arcs. Remind them to use a light, steady hand.

Activities: Art projects using geometric constructions; symmetry designs; Islamic geometric pattern drawing.

Remediation: Practise basic arc-drawing before formal constructions. Use a checklist for each construction.

Chapter 9: Symmetry

Learning Outcomes: Students identify lines of symmetry and orders of rotational symmetry; connect symmetry to regular polygons; explore symmetry in nature, art, and architecture.

Common Misconceptions: Students may believe any line through the centre is a line of symmetry. Emphasise: both halves must be mirror images.

Activities: Inkblot symmetry art; folding paper; symmetry in letters and rangoli.

Remediation: Use a mirror to test line symmetry. Physically rotate cut-out shapes to find rotational symmetry.

Chapter 10: The Other Side of Zero

Learning Outcomes: Students understand integers; compare and order on a number line; add, subtract, multiply, and divide integers; solve real-life integer problems.

Common Misconceptions: Students may think a larger-looking number (like -10) is greater than -2 . Use the number line constantly. For multiplication, they may forget the sign rules.

Activities: Temperature graphs; bank balance simulations; football score (positive goals/negative penalties).

Remediation: Reinforce number line placement daily. Use a sign-rules chart on the wall.

End of Grade 6 Mathematics Worksheet Book

Based on NCERT *Ganita Prakash* – 10 Chapters, 15 Worksheets

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